

QUANTUM PHYSICS

1. GENERAL

1.1 COURSE TITLE: Quantum Physics

1.2 *COURSE NUMBER: MC.PHY201.1

1.3 CONTACT HRS: 3-1-0, Credit 4

1.4 SEMESTER-OFFERED: Odd

1.5 PRE-REQUISITE: PHY-101: Classical, Quantum and Relativistic Mechanics

1.6 SYLLABUS COMMITTEE MEMBER: Dr. S. Upadhyay (Convener), Dr. S. K. Mishra, Prof. D. Giri

2. OBJECTIVE

To introduce the formal structure of Quantum Mechanics and few applications of Quantum Mechanics.

3. COURSE CONTENT

Review of wave mechanics: A brief review of foundations of Quantum Mechanics, WaveParticle duality, Heisenberg uncertainty principle, Wave function, Postulates of Quantum Mechanics, Schrodinger's Equation in one dimension. **(4 Lectures)**

Review of linear algebra: A brief overview of vector spaces matrices and vectors, eigenvalues and eigenvectors of matrices. **(6 Lectures)**

Operator method in Quantum mechanics: Hermitian Matrices, Linear operators, Dirac notation, Eigenvalue problem, Commutator Algebra, Functions of operators, Harmonic oscillator, Angular Momentum, Addition of angular momenta. **(10 Lectures)**

Schrodinger's Equation in three dimensions: Particle in a box problem in Cartesian and spherical polar coordinates, spherically symmetric potential, Hydrogen atom. **(5 Lectures)**

Identical Particles and Pauli Exclusion Principle. **(3 Lectures)**

Time Dependence in QM: Schrodinger, Heisenberg and Interaction Pictures. **(2 Lectures)**

Approximation methods in Quantum Mechanics: Variational method, Time independent perturbation theory and Time dependent perturbation theory. **(9 Lectures)**

4. READINGS

4.1 TEXTBOOK

1. Quantum Physics by S. Gasiorowicz, Publisher: Wiley; 3rd ed.

4.2 *REFERENCE BOOKS

1. Introduction to Quantum Mechanics by D J Griffiths, Publisher: Addison Wesley; 2 edition

2. Quantum Mechanics by C. Cohen-Tannoudji, Bernard Diu and F. Laloe, Publisher: Wiley-VCH;

3. Principles of Quantum Mechanics by R. Shankar, Publisher: Plenum Press; 2nd edition

5. OUTCOME OF THE COURSE

After completing this course the students are expected to have a fair knowledge of quantum states, operators and quantum measurements. Students will have a good practice for solving various problems in quantum mechanics and application of approximation methods. This course will set a basis for Atomic Physics, Condensed Matter, Statistical Mechanics courses. The course will also be helpful in realizing vector spaces and in shedding light on the usefulness of linear algebra for physical problems.